MCSC-202 • 3 CREDITS

Probability and Statistics

Model uncertainty, simulate systems, and make defensible statistical inferences from data.

STUDY BLOCKS	FORMAT	SCOPE
8	Definitions Equations Worked examples	Supplied syllabus only

How to use this handbook

For every study block: define the terms, reproduce the relation and its assumptions, solve the example without looking, and explain when the method is unsuitable.

Scope note: This handbook expands only topics named in the supplied syllabus. Confirm examination guidance, prescribed texts, and institutional policy with the responsible instructor.

Course map

1. Events and probability rules

Introduction, Events and outcomes, Probability rules, Conditional probability, Independence, Bayes' rule, Combinatorics

2. Random variables and discrete distributions

Random variables and distributions, Discrete random variables, Bernoulli, Binomial, Geometric, Negative binomial, Poisson

3. Continuous distributions, expectation and CLT

Probability densities, Exponential, Gamma, Weibull, Hyperexponential, Normal, Expectation and moments, Central Limit Theorem

4. Simulation and Monte Carlo

Simulation of discrete variables, Simulation of continuous variables, Monte Carlo methods, Performance evaluation, Bootstrapping

5. Stochastic processes

Concepts and classification, Bernoulli process, Poisson process, Markov chains, Simple queuing systems

6. Sampling, estimation and confidence intervals

Sample mean distribution, Sample proportion distribution, Parameters and statistics, Method of moments, Maximum likelihood, Pivotal quantity confidence intervals

7. Hypothesis testing and anomalies

Hypothesis testing, Type I and Type II errors, Anomalous events, Parameter estimation and testing

8. Exploration and statistical models

Graphics and exploratory data analysis, Simple and multiple regression, Least squares, Logistic regression, Analysis of variance, Robust statistics, Nonparametric statistics

Study block 01

Events and probability rules

Introduction	Events and outcomes
Probability rules	Conditional probability
Independence	Bayes' rule
Combinatorics	

Definition and idea

A sample space contains possible outcomes and an event is a subset. Conditional probability updates the relevant sample space. Independence means one event does not change the probability of the other. Bayes' rule reverses a conditional probability using priors and evidence.

```
P(A|B)=P(A intersection B)/P(B); P(A|B)=P(B|A)P(A)/P(B)
```

Worked example

If a detector is 95% sensitive, has 5% false positives, and prevalence is 1%, then a positive result has probability 0.0095/(0.0095+0.0495) about 16.1% of being a true case.

MUST UNDERSTAND
Always define the sample space and distinguish P(A|B) from P(B|A).

Study block 02

Random variables and discrete distributions

Random variables and distributions	Discrete random variables
Bernoulli	Binomial
Geometric	Negative binomial
Poisson	

Definition and idea

A random variable maps outcomes to numbers. A probability mass function assigns probability to each discrete value. Each named distribution encodes a particular experiment and assumptions.

```
Binomial: P(X=k)=C(n,k)p^k(1-p)^(n-k); Poisson: P(X=k)=e^(-lambda)lambda^k/k!
```

Worked example

Ten independent requests each fail with probability 0.1. The number of failures is Binomial(10,0.1), so P(no failures)=0.9^10.

MUST UNDERSTAND
Match the story to assumptions: fixed trials, waiting time, or event count.

Study block 03

Continuous distributions, expectation and CLT

Probability densities	Exponential
Gamma	Weibull
Hyperexponential	Normal
Expectation and moments	Central Limit Theorem

Definition and idea

A continuous density integrates to probability. Expectation summarizes a distribution's center; moments describe shape. The Central Limit Theorem explains why standardized sums or means often approach a normal distribution under suitable conditions.

```
E[X]=integral x f(x) dx; Z=(Xbar-mu)/(sigma/sqrt(n))
```

Worked example

If service times are exponential with rate 0.5 per minute, the mean is 2 minutes and P(X>4)=e^(-2). The CLT can approximate a mean of many independent service times.

MUST UNDERSTAND
A density value is not itself a probability; integrate over an interval.

Study block 04

Simulation and Monte Carlo

Simulation of discrete variables	Simulation of continuous variables
Monte Carlo methods	Performance evaluation
Bootstrapping	

Definition and idea

Simulation generates pseudo-random observations from a model. Monte Carlo estimates quantities by repeated sampling. Bootstrapping resamples observed data to approximate sampling variability when analytic formulas are difficult.

```
Monte Carlo estimate = (1/N) sum(g(X_i))
```

Worked example

Estimate pi by sampling N points uniformly in a unit square and multiplying the fraction inside the quarter circle by 4. Precision improves roughly with the square root of N.

MUST UNDERSTAND
Report seed, number of runs, uncertainty, and convergence evidence.

Study block 05

Stochastic processes

Concepts and classification	Bernoulli process
Poisson process	Markov chains
Simple queuing systems	

Definition and idea

A stochastic process is a time-indexed family of random variables. Markov chains condition the future on the current state. Queuing systems combine arrival and service processes to study waiting and congestion.

```
Markov: P(X_(t+1)=j | history)=P(X_(t+1)=j | X_t=i)
```

Worked example

For a two-state up/down system, a transition matrix gives the chance of repair or failure per step. Multiplying a state distribution by the matrix advances it one step.

MUST UNDERSTAND
State the index set, state space, transition assumptions, and whether time is discrete or continuous.

Study block 06

Sampling, estimation and confidence intervals

Sample mean distribution	Sample proportion distribution
Parameters and statistics	Method of moments
Maximum likelihood	Pivotal quantity confidence intervals

Definition and idea

A parameter describes a population; a statistic is computed from a sample. Estimation uses data to infer parameters. Method of moments matches sample and population moments; maximum likelihood chooses parameters that make observed data most plausible.

```
Xbar=(1/n)sum X_i; 95% large-sample CI: Xbar +/- 1.96 s/sqrt(n)
```

Worked example

A sample mean of 52 with s=10 and n=100 has approximate 95% interval 52 +/- 1.96, assuming conditions for the approximation are reasonable.

MUST UNDERSTAND
Report estimator, assumptions, standard error, confidence level, and interpretation.

Study block 07

Hypothesis testing and anomalies

Hypothesis testing	Type I and Type II errors
Anomalous events	Parameter estimation and testing

Definition and idea

A hypothesis test measures how incompatible data are with a null model. Type I error rejects a true null; Type II error fails to reject a false null. Anomaly detection needs an explicit baseline and false-alarm trade-off.

```
power = 1 - beta; significance level = alpha
```

Worked example

At alpha=0.05, repeated testing across many independent metrics creates more opportunities for false positives, so multiplicity should be considered.

MUST UNDERSTAND
A p-value is not the probability that the null hypothesis is true.

Study block 08

Exploration and statistical models

Graphics and exploratory data analysis	Simple and multiple regression
Least squares	Logistic regression
Analysis of variance	Robust statistics
Nonparametric statistics	

Definition and idea

Exploratory graphics reveal distribution, relationships, and data quality. Least squares models a numeric response; logistic regression models log-odds for a binary response. ANOVA compares variation among and within groups. Robust and nonparametric methods reduce reliance on fragile assumptions.

```
least squares minimizes sum(y_i - yhat_i)^2; logit(p)=log(p/(1-p))
```

Worked example

Plot residuals after regression: curvature suggests missing structure, while widening spread suggests non-constant variance. A coefficient must be interpreted with units and other predictors held constant.

> **MUST UNDERSTAND**
> Inspect data and residuals before trusting a fitted model.

Revision checklist

Use this final pass before an assessment or interview:

- ■ I can define every listed syllabus term in plain language.
- ■ I can reproduce the central equations and state their assumptions.
- ■ I can complete each worked example and create one variation.
- ■ I can compare closely related methods and justify a choice.
- ■ I can name limitations, failure modes, and ethical or operational concerns where relevant.